

Force and laws of motion: Source questions

IIT-JEE · 2007 · Paper I · Q10 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Newton-law and observer-frame practice

Question type: Assertion-reason multiple choice

Source: IIT-JEE 2007 · Paper I · Q10

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

A block moving on a rough horizontal surface stops after losing its kinetic energy to friction. The same surface is then inclined at 30 degrees and the block is projected up the incline with the same initial speed. Judge these claims: (1) the energy

Representation

**dissipated by friction before stopping is smaller on the incline;
(2) the coefficient of friction becomes smaller when the plane is inclined.**

Conditions

- Same pair of surfaces is used.
- Initial speed is the same in both trials.
- Incline angle is 30 degrees.

Common wrong route

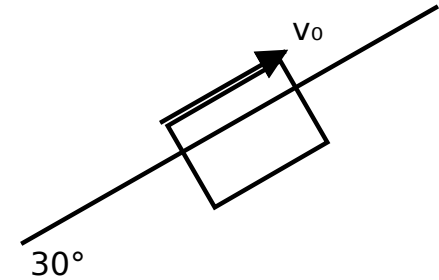
Assuming a smaller friction force means the coefficient itself has changed.

same block · same initial speed

level trial



30° incline trial



rough contact in both trials
block eventually stops

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

Need the concept again?

Quantitative static and kinetic friction conditions

IIT-JEE · 2008 · Paper II · Q33 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Newton-law and observer-frame practice

Question type: Assertion-reason multiple choice

Source: IIT-JEE 2008 · Paper II · Q33

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

Judge the pair of statements: (1) pulling a heavy body across level ground is easier than pushing it when the applied force is oblique; (2) frictional force depends on the nature of the surfaces in contact.

Representation

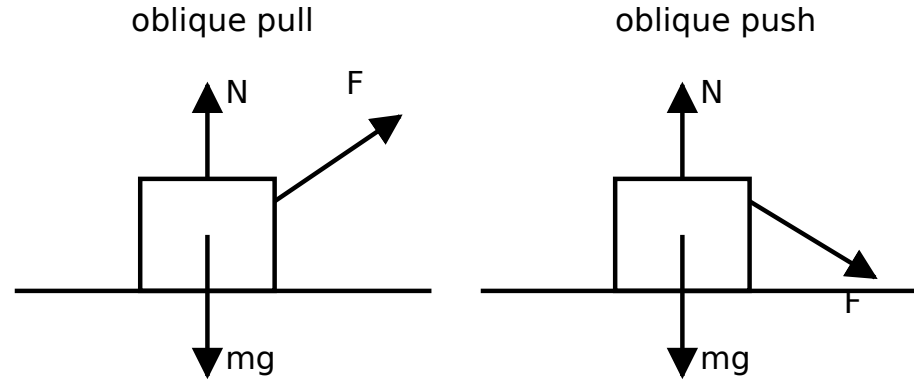
Conditions

- Compare oblique pull and oblique push on the same level surface.

Common wrong route

Using the true surface-nature statement as the causal explanation for pull versus push.

same body · same rough level surface



HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

Need the concept again?

Quantitative static and kinetic friction conditions

IIT-JEE · 2011 · Paper I · Q41 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Newton-law and observer-frame practice

Question type: Integer answer

Source: IIT-JEE 2011 · Paper I · Q41

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

A block is on a plane inclined at 45 degrees. The coefficient of friction is μ . The force needed to just push the block upward is three times the force needed to just prevent it from sliding downward. If $N = 10\mu$, find N .

Representation

Conditions

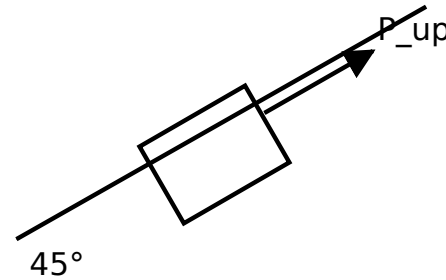
- Both forces act parallel to the incline.
- Each case is at limiting equilibrium.

Common wrong route

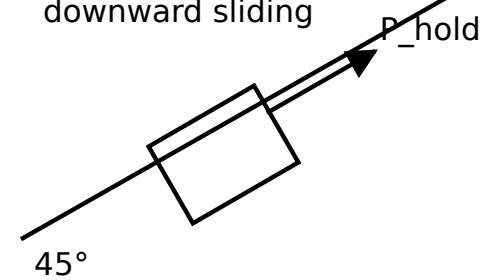
Using the same friction direction in both limiting cases.

45° rough incline

applied force parallel to plane
case A: just pushes upward



case B: prevents
downward sliding



coefficient = μ

both cases at limiting equilibrium

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

Need the concept again?

Quantitative static and kinetic friction conditions

JEE (Advanced) · 2014 · Paper 2 · Q19 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Newton-law and observer-frame practice

Question type: Matrix match

Source: JEE (Advanced) 2014 · Paper 2 · Q19

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

Two touching blocks $m_1=1$ kg and $m_2=2$ kg rest on an incline of angle θ , with m_1 placed just above m_2 (up the slope), so m_1 presses on m_2 . The plane is smooth under m_1 ; for m_2 , static and kinetic friction coefficients are

Representation

both 0.3. Determine which friction expression applies to m_2 at $\theta = 5^\circ, 10^\circ, 15^\circ$ and 20° . Candidate expressions are: (1) $m_2 g \sin \theta$; (2) $(m_1 + m_2)g \sin \theta$; (3) $0.3 m_2 g \cos \theta$; (4) $0.3(m_1 + m_2)g \cos \theta$.

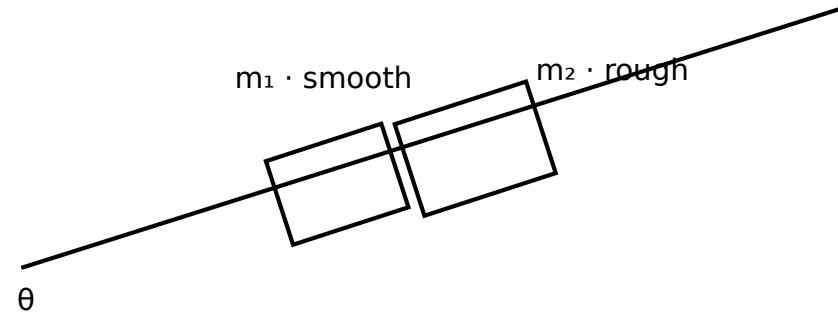
Conditions

- $m_1 = 1 \text{ kg}$; $m_2 = 2 \text{ kg}$.
- Only m_2 has friction with the plane, $\mu = 0.3$.
- Use $\tan 5.5^\circ \approx 0.1$, $\tan 11.5^\circ \approx 0.2$, $\tan 16.5^\circ \approx 0.3$.
- m_1 is directly up-slope of m_2 and pushes on it; m_2 is the lower block.

Common wrong route

Using $m_2 g \sin \theta$ as the required static friction and ignoring the contact

$m_1 = 1 \text{ kg}$ above $m_2 = 2 \text{ kg}$



$\mu = 0.3$ only under $m_2 \cdot \theta = 5^\circ, 10^\circ, 15^\circ, 20^\circ$

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

force transmitted by m_1 .

H3 Equation / formal model **Guided**

H4 Assembly checkpoint **Guided**

Need the concept again?

Quantitative static and kinetic friction conditions

JEE (Advanced) · 2020 · Paper 1 · Q13 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Newton-law and observer-frame practice

Question type: Numerical value

Source: JEE (Advanced) 2020 · Paper 1 · Q13

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

A uniform metre scale is supported horizontally on two index fingers initially at 0.00 cm

Representation

and 90.00 cm. Both fingers are moved slowly toward the centre. The slipping finger alternates because static and kinetic friction differ. With $\mu_s=0.40$ and $\mu_k=0.32$, find the final distance x_R of the right finger from the 50.00 cm centre at the instant it stops and the left finger starts slipping again.

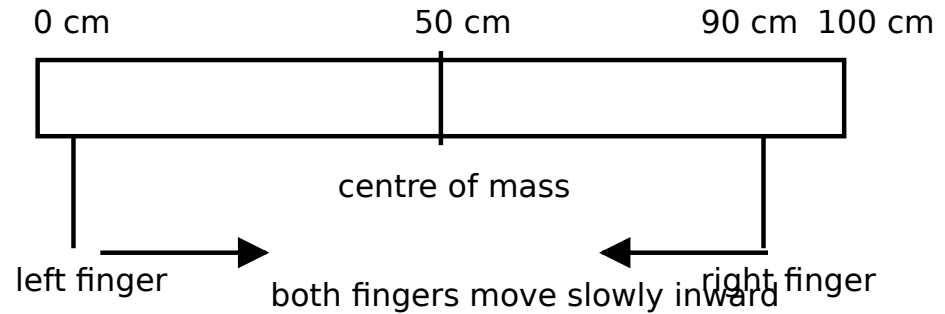
Conditions

- Scale is uniform.
- Finger motion is quasistatic.
- $\mu_s=0.40$; $\mu_k=0.32$.

Common wrong route

Using one fixed normal reaction or one friction state throughout both stages, so

uniform metre scale · $\mu_s = 0.40$ · $\mu_k = 0.32$



HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

H3 Equation / formal model Guided

the second stick-slip threshold is written with stale contact data.

H4 **Assembly checkpoint** **Guided**

Need the concept again?

Quantitative static and kinetic friction conditions

JEE Main · 2026 · Session 2 · 06 Apr 2026 · Shift 2 · Q46

Official past paper, faithful restatement of the original

Subject: Physics **Topic:** Laws of Motion **Concept:** Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept **Question difficulty:** Multi-step reasoning

Family: Newton-law and observer-frame practice **Question type:** Integer answer

Source: JEE Main 2026 · 06 Apr · Shift 2 · Q46 **Provenance:** Verified past-exam adaptation

ATTEMPT FIRST

A block takes time t to slide from rest down a 45° inclined plane. When the same plane is made frictionless, the time becomes $t/2$.

Representation

If the original coefficient of friction is $\alpha/100$, find α .

Conditions

- The block starts from rest in both cases and traverses the same distance along the plane.

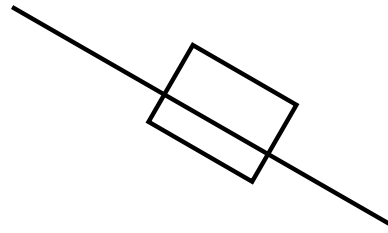
Common wrong route

Assuming halving the time only doubles acceleration, or omitting the normal-force factor in kinetic friction.

same 45° incline · same distance

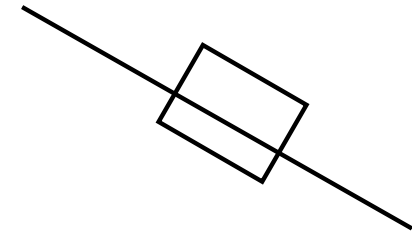
starts from rest
rough trial: time t

smooth trial: time $t/2$



rough contact

incline angle = 45°



frictionless

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

H3 Equation / formal model **Guided**

H4 Assembly checkpoint **Guided**

Need the concept again?

Quantitative static and kinetic friction conditions

INJSO · 2011 · 29 Jan 2011 · Q22 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Newton-law and observer-frame practice

Question type: Single-correct multiple choice

Source: INJSO 2011 · Q22

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

**A 5 kg block is to be dragged
along a rough horizontal surface
with coefficients $\mu_s=0.5$ and**

Representation

$\mu_k=0.3$. A horizontal force of 20 N is applied. What are the block's acceleration and the friction force?

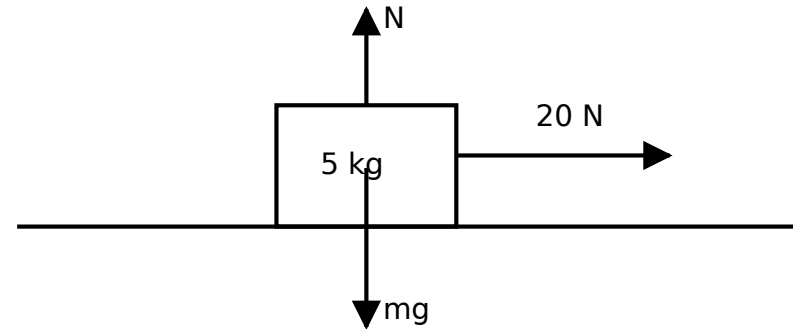
Conditions

- The surface is horizontal.
- The block begins in the no-slip state.
- Use the paper's stated $g=10 \text{ m/s}^2$.

Common wrong route

Taking static friction to be $\mu_s N$ merely because μ_s is given, instead of finding the friction required for rest.

$$m = 5 \text{ kg} \cdot \mu_s = 0.5 \cdot \mu_k = 0.3$$



initially no slip; decide the contact state first

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

Need the concept again?

Quantitative static and kinetic friction conditions

INJSO · 2016 · 30 Jan 2016 · Q32A Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Newton-law and observer-frame practice

Question type: Constructed response

Source: INJSO 2016 · Q32A

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

Two solid cubical boxes P (4.0 kg) and Q (8.0 kg) stay in contact while a horizontal force F on P pushes them along a rough horizontal table at constant speed. Their kinetic-friction coefficients with the table are 0.4

Representation

and 0.6 respectively. Find F ; determine the magnitude and direction of the resultant force exerted by the table on Q ; and give the labelled free-body force account for both boxes.

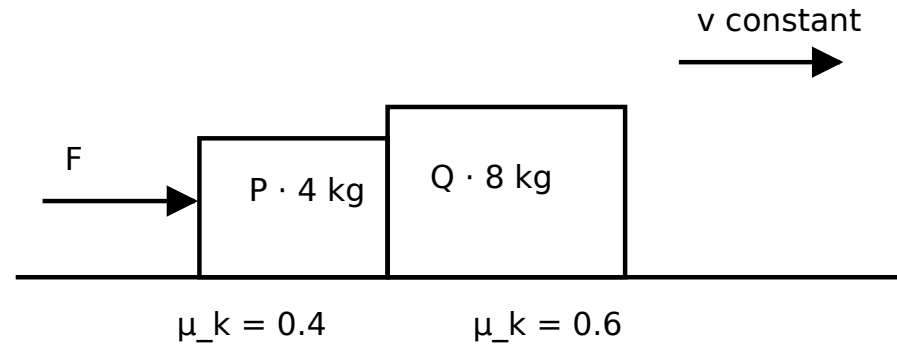
Conditions

- P and Q remain in contact and move at constant speed.
- $\mu_k(P\text{-table})=0.4$ and $\mu_k(Q\text{-table})=0.6$.
- Use the paper's stated $g=10\text{ m/s}^2$.
- The boxes have the same density, as stated in the source.

Common wrong route

Using only one friction coefficient for the two-box system, or treating the P -

constant speed right · boxes remain in contact



HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

on-Q contact force as part of the table's resultant reaction on Q.

H3 Equation / formal model Guided

H4 Assembly checkpoint Guided

Need the concept again?

Quantitative static and kinetic friction conditions

INJSO · 2018 · 27 Jan 2018 · Q12 Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Newton-law and observer-frame practice

Question type: Single-correct multiple choice

Source: INJSO 2018 · Q12

Provenance: Verified past-exam adaptation

ATTEMPT FIRST

A 100 kg block is on a level rough surface. One person pushing with force F cannot start it. With three

Representation

people, each pushing with the same force F , it moves with acceleration 0.5 m/s^2 . After one person stops pushing, the block moves with uniform speed. Find F and the coefficient of kinetic friction μ .

Conditions

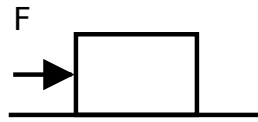
- Mass=100 kg.
- The surface is level.
- Three equal pushes produce 0.5 m/s^2 acceleration; two equal pushes produce uniform speed while moving.
- Use the paper's stated $g=10 \text{ m/s}^2$.

Common wrong route

Trying to infer μ from the one-person failure to move, or changing the kinetic-

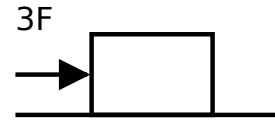
same 100 kg block · level rough surface

μ unknown
1 person



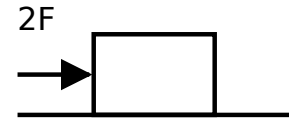
cannot start

3 people



$a = 0.5 \text{ m/s}^2$

2 people



uniform speed

Last two trials occur while the block is moving.

Find F and the kinetic-friction coefficient.

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

friction force between the two moving trials.

H3 Equation / formal model Guided

H4 Assembly checkpoint Guided

Need the concept again?

Quantitative static and kinetic friction conditions

INJSO · 2023 · 28 Jan 2023 · Q24 Official past paper, faithful restatement of the original

Subject: Physics **Topic:** Laws of Motion **Concept:** Friction opposes relative sliding at a contact

Concept difficulty: Medium concept **Question difficulty:** Connected reasoning

Family: Newton-law and observer-frame practice **Question type:** Multiple-correct multiple choice

Source: INJSO 2023 · Q24 **Provenance:** Verified past-exam adaptation

ATTEMPT FIRST

A rider is cycling on a level road, applies a brake, and the cycle slows. Select the correct

Representation

statements about which agent supplies the force that reduces the cycle's speed when only the front brake or only the rear brake is used.

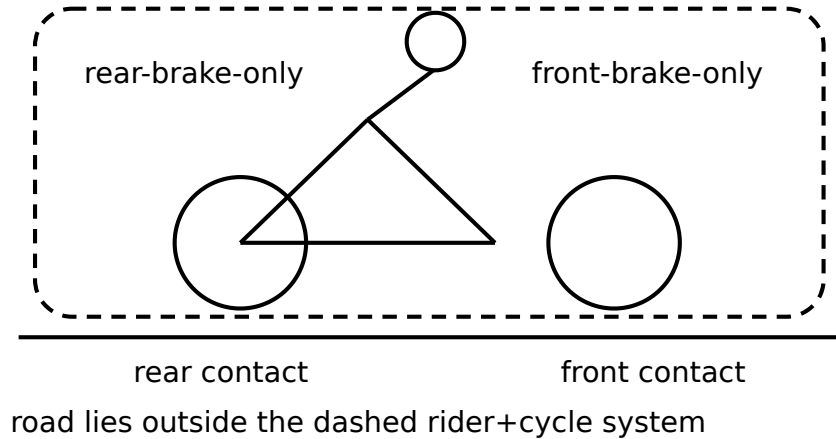
Conditions

- The cycle travels on a level road.
- Consider the front-brake-only and rear-brake-only cases separately.

Common wrong route

Naming a tyre as the external agent that slows the whole rider+cycle system because the brake acts on that tyre.

system boundary: rider + whole cycle



HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Representation Guided

H1 Key concept Guided

H2 Crux Guided

Need the concept again?

Friction opposes relative sliding at a contact

Physics · Force and laws of motion